Product Evaluation — Live Translate

- **Student:** Amina Javaid
- **Date:** 2026-07-13
- **Video demo:** [FDE Live Translate Demo]
- **LLM provider / model:** OpenAI / `gpt-4o-mini`
- **Backend target:** benchmark run locally at `http://localhost:8787`; live-website test against the deployed gateway at `https://fde-translate-gw-aj.fly.dev` (AI service private at `fde-translate-ai-aj.internal:8000`)

Verdict

> This is shippable as an MVP. The product does exactly what it promises: the provided widget lights up on a real, third-party site and flips the page into fluent Mexican Spanish, with prices, SKUs, and model codes left intact. The strongest part is the caching layer — a two-tier (memory + SQLite) cache that survives restarts, delivers a ~56× speedup on repeats (hit p95 23 ms vs miss p95 1299 ms), holds a 75% hit rate under load with **zero errors**, and cuts projected monthly LLM spend by ~75%. Correctness is handled the right way: on an LLM error the service **fails loud** (HTTP 5xx) rather than silently serving English, and a single request ID is traceable across both services' logs. The weakest points are operational, not functional: cache-miss latency (~1.3 s p95) is inherited from the LLM provider and is the main user-visible wait; the cost figures below use the repo's placeholder Sonnet pricing, so real `gpt-4o-mini` cost is materially lower; and page coverage of lazily/dynamically-loaded content on very large sites was not exhaustively audited.

Rubric score (from `eval/report.json`): 70 / 70 auto (+ 30 manual)

Auto breakdown: widget_lights_up 15/15 · caching_correctness 20/20 · performance_sla 15/15 · logging_observability 10/10 · service_separation_contract 10/10.

1. Performance & cost (from `benchmark/bench.py`, clean cold→warm run)

Metric	Result	SLA	Pass?
Cache hit p95	23.4 ms	≤ 60 ms	✅
Cache miss p95	1299 ms	≤ 3500 ms	✅
Cache hit rate	75.0 %	≥ 60 %	✅
Throughput	703.5 req/s	≥ 20	✅

Metric	Result	SLA	Pass?
Error rate	0.0 %	≤ 1 %	✅
Cost per miss	$0.000157	—	—
Monthly savings from cache	$58.95	—	—

SLA gate: ✅ **ALL SLAs MET.** Cache miss p50/p95 = 1070/1299 ms; cache hit p50/p95 = 4/23 ms (≈56× speedup).

> **Cost caveat:** `benchmark/sla.json` ships with **placeholder Anthropic** `claude-sonnet-4-6` **pricing ($3/$15 per Mtok)**. This product actually runs on OpenAI `gpt-4o-mini` (~$0.15/$0.60 per Mtok), so real cost is roughly **15–20× lower** than the numbers above — i.e. genuine monthly savings are larger in absolute cache-avoided calls but smaller in dollar terms. `sla.json` is on the do-not-edit list, so it was left unchanged; interpret the dollar figures as an upper bound.

2. Live-website test

- **Site tested:** https://www.homedepot.com (real, third-party, content-rich site)
- **Translated whole page?** Yes — loaded via the Chrome extension (Load unpacked → `extension/`) pointed at the production gateway `https://fde-translate-gw-aj.fly.dev`. Clicking **Translate page** flipped the visible page into Mexican Spanish with layout intact.
- **Coverage gaps:** Static page text translated cleanly. Lazily/dynamically-loaded content that renders after the translate pass (e.g. content revealed on scroll) and text baked into images are not covered — expected for a one-shot DOM pass.
- **Cache on re-translate:** Restore → Translate again returns instantly with `cached:true`. Verified directly against production: first call `cached:false` ~3.0 s, identical repeat `cached:true` 0 ms.
- **Resilience:** The **console-snippet** loader is blocked by Home Depot's strict Content-Security-Policy (expected, and documented in the brief) — the **Chrome extension** path is used instead and works, because it injects via the extension context and proxies through the gateway. No layout breakage; page remained usable.
- **Screenshots:** captured live by the student during the demo — attach before/after to the submission (also shown in the video).

Sample translations (6–8) — real output from the deployed product

Original (EN)	Translation (es-MX)	Numbers/prices/codes kept?	OK?
Add to Cart	Agregar al carrito	n/a	✅ (es-MX "agregar", not Castilian "añadir")
Free delivery on orders over $45.00	Envío gratis en pedidos mayores a $45.00	$45.00 ✅	✅
In stock at your store — 12 available (SKU 1001234567)	En stock en tu tienda — 12 disponibles (SKU 1001234567)	12, SKU 1001234567 ✅	✅ (informal "tu" = es-MX)
Milwaukee M18 FUEL 18-Volt Cordless Drill, model 2904-20 — $199.00	Taladro inalámbrico Milwaukee M18 FUEL de 18 voltios, modelo 2904-20 — $199.00	model 2904-20, $199.00 ✅	✅
Sign in for faster checkout and order tracking.	Inicia sesión para un pago más rápido y seguimiento de pedidos.	n/a	✅
4.5 out of 5 stars (2,341 reviews)	4.5 de 5 estrellas (2,341 reseñas)	4.5, 5, 2,341 ✅	✅
Buy 2, save 15% on select power tools.	Compra 2, ahorra 15% en herramientas eléctricas seleccionadas.	2, 15% ✅	✅
Estimated arrival: Tuesday, July 21	Fecha estimada de llegada: martes 21 de julio	21 ✅	✅

3. Dimension scorecard

Dimension	Pass / Partial / Fail	Evidence
Translation accuracy	Pass	8/8 samples fluent and correct; matches source meaning
Mexican-Spanish register (es-MX)	Pass	"agregar al carrito" (not Castilian "añadir"), informal "tu tienda"; prompt pins es-MX explicitly
Numbers / prices / codes preserved	Pass	$45.00, $199.00, model 2904-20, SKU 1001234567, 15%, 2,341 all preserved verbatim
Page coverage	Pass (with note)	Full static page translated on homedepot.com; dynamic/lazy content and image text not covered

Dimension	Pass / Partial / Fail	Evidence
Cache effectiveness	Pass	75% hit rate, 56× speedup, survives restart (disk hit after reboot), production repeat = 0 ms
Latency vs SLA	Pass	All 5 SLA thresholds met; hit p95 23 ms, miss p95 1299 ms
Error handling (no silent English)	Pass	LLM errors propagate as HTTP 5xx (verified); no try/except returning source text
Resilience on a real site	Pass	Works via extension on strict-CSP site; console-injection CSP block handled by using the extension path
UX polish	Pass	Provided widget unmodified; backend URL configurable in extension popup; `/health` reports nested AI status

4. Top fixes before shipping

1. **Correct the cost model** — update `benchmark/sla.json` `cost_model` to OpenAI `gpt-4o-mini` published rates so the reported dollar figures reflect the real provider (currently placeholder Sonnet pricing).
2. **Broaden dynamic-content coverage** — re-translate on DOM mutations (e.g. a `MutationObserver`) so content that loads after the initial pass (infinite scroll, tabs) is also translated.
3. **Production hardening (stretch goals)** — add rate limiting (429), a cache TTL + `POST /clear-cache` endpoint, and a GitHub Action for automated Fly deploys. Cache-miss latency (~1.3 s) could be masked with response streaming.

Deployment

- **Public gateway:** https://fde-translate-gw-aj.fly.dev (HTTPS, 2 machines, HA)
- **AI service:** private on Fly's internal network (`fde-translate-ai-aj.internal:8000`), no public IP — the OpenAI key lives only as a Fly secret, never at the browser edge
- **Source:** https://github.com/amina-javaid/fde-live-translate
- Both services deploy from Docker (`Dockerfile.ai`, `Dockerfile.gateway`) via `fly.ai.toml` / `fly.gateway.toml`; SQLite cache persisted on a Fly volume.